





Q



Q





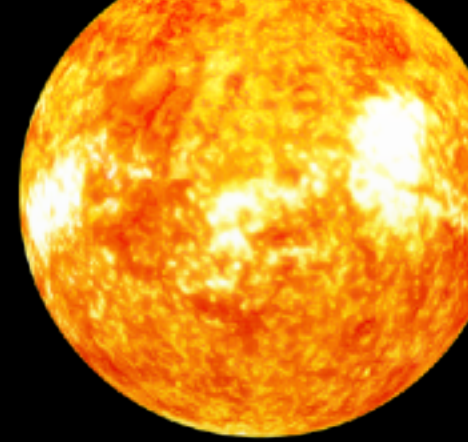






Solar Neutrino Puzzle

Solar Neutrino



☀ Why does Sun shine?

— Each chain of nuclear reactions



— releases $Q=26.7$ MeV of energy.

☀ Let us estimate neutrino flux at Earth:

— number of neutrino per second = 2 x number of chains per second

— Sun luminosity = Q x number of chains per second.

☀ Let us divide left and right parts:

$$\frac{\text{number of neutrino per sec}}{\text{Sun luminosity}} = \frac{2}{Q}$$

☀ Divide numerator and denominator by R^2 :

$$\frac{\text{number of neutrino per sec}/R^2}{\text{Sun luminosity}/R^2} = \frac{2}{Q}$$

☀ Insert numbers. Flux of solar energy at Earth orbit (Sun luminosity/ R^2) = 1.4 kW/m^2 :

$$\begin{aligned}\text{Neutrino flux at Earth} &= \\ &= \text{number of neutrino per sec}/R^2 \\ &= 6.5 \cdot 10^{10} / \text{cm}^2 / \text{sec}\end{aligned}$$

Solar Neutrino Puzzle

😲 Experiments discovered two times smaller flux of neutrino!

